Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 1

Time Allowed: 30 Minutes

General Instructions:

Read the following instructions carefully.

1. There are 16 questions in this question paper with internal choice.
2. SECTION A consists of 16 multiple-choice questions carrying 1 mark each.
3. **All questions are compulsory.**
4. **Use of log tables and calculators is not allowed.**

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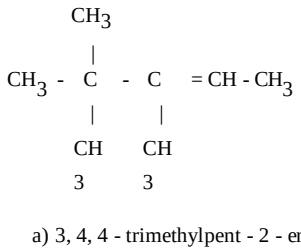
Section A

1. Name the following compound as per the IUPAC system **[1]**



a) 3, 4, 4 - trimethylpent - 2 - ene
b) 2, 2, 3 - trimethylpent - 4 - ene
c) 4 diethyl, 2 - ethyl pentene
d) 2 diethyl, 3 - ethyl pentene

2. On hydrolysis, which of the following carbohydrates gives only glucose? **[1]**

a) Lactose
b) Starch
c) Sucrose
d) Fructose

3. When nitrobenzene is heated with tin and concentrated HCl, the product formed is: **[1]**

a)
b)
c)
d)

4. What kind of compounds undergo Cannizaro reactions? **[1]**

a) Aldehydes with no α- hydroge
b) Ketones with no α- hydrogen
c) Carboxylic acids with α- hydrogen
d) Aldehydes with α- hydrogen

5. The rate law for a particular reaction is given as rate = $k[A][B]^2$. **[1]**

How is the rate of reaction affected if we double the concentration of B?

a) becomes half $(\frac{1}{2})$
b) four times
c) three times
d) two times

6. Match the types of solutions given Column I to the examples given in Column II. **[1]**

Column I	Column II
(a) Solution of gas in gas	(i) Brass
(b) Solution of solid in solid	(ii) Air
(c) Solution of liquid in gas	(iii) Zinc amalgam
(d) Solution of liquid in solid	(iv) Chloroform in Nitrogen

a) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
b) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)
c) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)
d) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

7. Carbon tetrachloride has a dipole moment: **[1]**

a) $\mu = 0$
b) $\mu = 1$
c) $\mu = 4$
d) $\mu = 2$

8. The interstitial compounds of transition metals are **[1]**

a) more ductile
b) softer
c) harder
d) more metallic

9. Consider the reaction **[1]**

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

The equality relation between $\frac{d[NH_3]}{dt}$ and $\frac{-d[H_2]}{dt}$ is:

a) $\frac{d[NH_3]}{dt} = -\frac{3}{2}\frac{d[H_2]}{dt}$
b) $\frac{d[NH_3]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt}$
c) $\frac{d[NH_3]}{dt} = -\frac{2}{3}\frac{d[H_2]}{dt}$
d) $\frac{d[NH_3]}{dt} = -\frac{d[H_2]}{dt}$

10. Benzaldehyde and acetone can be best distinguished by using: **[1]**

a) Sodium hydroxide solution
b) 2, 4 – DNP reagent
c) Hydrazine
d) Tollen's reagent

11. Williamson's synthesis is used for the preparation of **[1]**

a) ethers
b) alkyl halides
c) aldehydes
d) alcohols

12. Which of the following would not be a good choice for reducing nitrobenzene to aniline? **[1]**

a) Sn and HCl
b) LiA/H_4
c) H_2/Hi
d) Fe and HCl

13. **Assertion (A):** Vitamin C cannot be stored in our body. **[1]**

Reason (R): Vitamin C is water soluble and readily excreted in urine.

a) Both A and R are true and R is the correct
b) Both A and R are true but R is not the

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explanation of A.

correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

14. **Assertion (A):** Isobutanal does not give the iodoform test. **[1]**

Reason (R): It does not have alpha-hydrogen.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

15. **Assertion (A):** In monohaloarenes, further electrophilic substitution occurs at ortho and para positions. **[1]**

Reason (R): Halogen atom is a ring deactivator.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

16. **Assertion (A):** The boiling point of ethanol is higher than that of dimethyl ether. **[1]**

Reason (R): Ethanol molecules are associated through hydrogen bonding whereas in dimethyl ether, it is not possible.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

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